Lecture Notes · Arithmetic · Session 1 — Divisibility, Euclidean division and the Euclidean algorithm

All the session's material with complete proofs: division with remainder (existence via well-ordering, and uniqueness), the lemma that justifies the Euclidean algorithm, and Bézout's identity by its two routes — the structural one and the extended algorithm with a fully worked example. Exercises and a Going further section at the end.

1.1 Divisibility

Definition 1.1.1. For $a, b \in \mathbb{Z}$: a **divides** b, written $a \mid b$, if there exists $c \in \mathbb{Z}$ with $b = ac$.

Proposition 1.1.2. For arbitrary integers: 1. *(Transitivity)* $a \mid b$ and $b \mid c \Rightarrow a \mid c$. 2. *(Combinations)* if $a \mid b$ and $a \mid c$, then $a \mid (xb + yc)$ for all $x, y \in \mathbb{Z}$. 3. If $a \mid b$ and $b \neq 0$, then $|a| \leq |b|$.

Proof. (1) $b = au$, $c = bv \Rightarrow c = a(uv)$. (2) $b = au$, $c = av \Rightarrow xb + yc = a(xu + yv)$. (3) $b = ac$ with $c \neq 0$, so $|c| \geq 1$ and $|b| = |a||c| \geq |a|$. ■

> Property (2), so modest, is the key to half the session: Bézout lives off it.

1.2 Euclidean division

Theorem 1.1.3 (division with remainder). For $a \in \mathbb{Z}$ and $b > 0$ there exist **unique** integers q (quotient) and r (remainder) with

$$a = bq + r, \qquad 0 \leq r < b.$$

Proof. **Existence.** The set $S = \{\, a - bq : q \in \mathbb{Z},\ a - bq \geq 0 \,\}$ is nonempty (for q very negative, $a - bq$ becomes positive). By the **well-ordering principle** (Proposition 0.1.9: every nonempty subset of $\mathbb{N}$ has a least element), S has a least element $r = a - bq_0 \geq 0$. We claim $r < b$: if we had $r \geq b$, then $r - b = a - b(q_0 + 1) \geq 0$ would be in S and smaller than r, against minimality.

Uniqueness (the "subtract and bound" trick). If $a = bq + r = bq' + r'$ with $0 \leq r, r' < b$, subtracting: $b(q - q') = r' - r$. The right-hand side satisfies $|r' - r| < b$; the left-hand side is a multiple of b. The only multiple of b with absolute value less than b is 0: hence $r = r'$ and, since $b \neq 0$, $q = q'$. ■

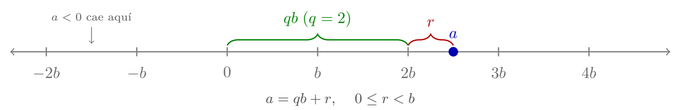

Figure 1: Euclidean division on the number line: a falls between two consecutive multiples of b; the quotient q counts them and the remainder r is what is left over ($0 \leq r < b$). The marks continue **in both directions**: when $a < 0$ it falls to the left of the origin, and the mark to its left is the one giving the quotient — this is where the convention of Remark 1.1.4 comes from.

Remark 1.1.4 (the convention matters — and it *is* a convention). For $a = -7$, $b = 3$: $-7 = 3 \cdot (-3) + 2$ — the remainder is 2, not -1. The convention $0 \leq r < b$ makes the expression unique and is the one congruences use (Session 3).

So why do we require $b > 0$? For convenience, not out of necessity. The general version holds for every $b \neq 0$: there exist **unique** q, r with $a = bq + r$ and $0 \leq r < |b|$. And it reduces to the positive case in one line: apply the theorem to $|b|$, which gives $a = |b|q' + r$; if $b < 0$ then $|b| = -b$, so $a = b(-q') + r$ and it suffices to take $q = -q'$. For instance, $7 = (-3) \cdot (-2) + 1$, with $0 \leq 1 < 3 = |-3|$. What we **do** have to exclude is $b = 0$: then $a = 0 \cdot q + r$ forces $r = a$, and the condition $0 \leq r < 0$ is impossible to satisfy — it is division by zero, and the **only** genuine exclusion in the theorem. We state it with $b > 0$ because in everything that follows the divisor is positive by nature (the modulus n of congruences, the remainders in Euclid's algorithm, the gcd normalised to be ≥ 0), which spares the later statements a forest of absolute-value bars.

> *(In $K[x]$ —Polynomials module— the analogous hypothesis is precisely* ***"nonzero divisor"****: $g \neq 0$ with $\deg r < \deg g$. There is no order there letting us ask for "positive", only the degree; the dictionary is $|\cdot| \leftrightarrow \deg$. Stated with $b \neq 0$ and $|b|$, division in $\mathbb{Z}$ and division in $K[x]$ say literally the same thing: nonzero divisor, remainder smaller than the divisor in the measure proper to each world.)*

And uniqueness depends on the band, not on the sign of b. Fixing $0 \leq r < |b|$ is **one** choice among several, and all of them give existence and uniqueness: the remainder carrying the sign of the **dividend** (what C, C++, Java and Rust do, where `-7 % 3` is -1), or the **centred** remainder, $-|b|/2 < r \leq |b|/2$, convenient in cryptography and in lattice reduction. Ours is **Python**'s convention: `-7 % 3` returns `2`. It is worth checking in the interpreter, because that disagreement between languages is a classic source of bugs.

> *Structural lens (a mention):* $\mathbb{Z}$ is the model of a world "with division with remainder". This single property generates everything that follows — and in the Polynomials module we will see that $K[x]$ shares it, with the **degree** in the role of the absolute value.

1.3 Greatest common divisor and the Euclidean algorithm

Definition 1.1.5. For a, b not both zero, $\gcd(a, b)$ is the largest of the common divisors. (It exists: the common divisors are bounded by 1.1.2.3 and the set is nonempty — it contains 1.) When $\gcd(a, b) = 1$ we say that a and b are **coprime** (or *relatively prime*): their only positive common divisor is 1, that is, they share no factor greater than 1. This is the vocabulary in which Euclid's lemma (Session 2), the invertibility criterion (Session 3) and the Chinese remainder theorem (Session 4) are stated.

> **On "the largest" and the sign.** Divisors come in pairs $\pm d$, so the largest in the usual order of $\mathbb{Z}$ is the **positive** one: thus the gcd is an integer ≥ 0, well defined and **unique** — "the largest" is literal. The characterization by divisibility (d such that every common divisor divides d; exercise 9) determines it only **up to sign**, and the convention fixes $\gcd \geq 0$. (The case $a = b = 0$ is excluded on purpose: **every** integer divides 0, so there is no greatest common divisor.) *(In $K[x]$ — the Polynomials module — the gcd is unique up to a nonzero constant factor, and it is normalized by making it monic: the same "up to a unit" issue.)*

Lemma 1.1.6 (why the Euclidean algorithm works). If $a = bq + r$, then the **common divisors of** (a, b) **are exactly the common divisors of** (b, r). In particular, $\gcd(a, b) = \gcd(b, r)$.

Proof. If $d \mid a$ and $d \mid b$: since $r = a - bq$ is a combination of a and b, $d \mid r$ (Prop. 1.1.2.2) — a common divisor of (b, r). Conversely, if $d \mid b$ and $d \mid r$: $a = bq + r$ is a combination of b and r, so $d \mid a$. The two pairs have *the same set* of common divisors; in particular, the same maximum. ■

Algorithm 1.1.7 (Euclid). Divide a by b, replace $(a, b) \mapsto (b, r)$, repeat until the remainder is 0; the gcd is **the last nonzero remainder**.

Justification. Each step preserves the gcd (Lemma 1.1.6). The remainders form a strictly decreasing sequence of nonnegative integers ($0 \leq r < b$), which by well-ordering (Proposition 0.1.9) **cannot be infinite**: the algorithm terminates. On reaching $(d, 0)$, $\gcd(d, 0) = d$. ■

> **Example 1.1.8.** $\gcd(252, 198)$:
>
> $$252 = 1 \cdot 198 + 54 \qquad 198 = 3 \cdot 54 + 36 \qquad 54 = 1 \cdot 36 + 18 \qquad 36 = 2 \cdot 18 + 0$$
>
> $\gcd(252, 198) = 18$.

1.4 Bézout's identity

Theorem 1.1.9 (Bézout). For a, b **not both zero**, there exist $x, y \in \mathbb{Z}$ such that $\gcd(a, b) = ax + by$.

Proof (structural). Let $D = \{\, ax + by : x, y \in \mathbb{Z} \,\}$ and let d be the **least positive element** of D (it exists: since a, b are not both zero, D contains some positive element — if $a \neq 0$, $|a| \in D$; if $a = 0$, then $b \neq 0$ and $|b| \in D$; well-ordering, Proposition 0.1.9). We claim that $d = \gcd(a, b)$:

- d **divides** a**:** dividing, $a = dq + r$ with $0 \leq r < d$; then $r = a - dq$ is a combination of a and b (because d is one), that is, $r \in D$ with $0 \leq r < d$. By minimality of d, $r = 0$. Likewise for b.
- **It is the largest:** any common divisor c of a and b divides every combination $ax + by$ (Prop. 1.1.2.2), in particular d; since $d > 0$, this forces $|c| \leq d$, and therefore $c \leq d$. Thus d is the largest of the common divisors. ■

Method 1.1.10 (extended Euclidean algorithm). The coefficients x, y are **constructed** by working back up the divisions of Example 1.1.8, solving for each remainder:

$$18 = 54 - 36 = 54 - (198 - 3 \cdot 54) = 4 \cdot 54 - 198$$

$$= 4(252 - 198) - 198 = \mathbf{4 \cdot 252 - 5 \cdot 198}.$$

Check: $4 \cdot 252 - 5 \cdot 198 = 1008 - 990 = 18$ ✓.

Remark 1.1.11 (the coefficients are not unique — and which ones are *all* of them). Method 1.1.10 hands you **one** pair (x, y), and there are **infinitely many**. Fixing $d = \gcd(a, b)$ and a pair (x_0, y_0) with $ax_0 + by_0 = d$, the complete family is

$$x = x_0 + k\,\tfrac{b}{d}, \qquad y = y_0 - k\,\tfrac{a}{d}, \qquad k \in \mathbb{Z}.$$

They all work (a direct computation — the k terms cancel):

$$a\left(x_0 + k\tfrac{b}{d}\right) + b\left(y_0 - k\tfrac{a}{d}\right) = ax_0 + by_0 + k\left(\tfrac{ab}{d} - \tfrac{ab}{d}\right) = d.$$

And there are no others. If $ax + by = d$, subtracting from $ax_0 + by_0 = d$ leaves $a(x - x_0) = -b(y - y_0)$; dividing by d,

$$\tfrac{a}{d}(x - x_0) = -\tfrac{b}{d}(y - y_0).$$

So $\frac{b}{d}$ divides $\frac{a}{d}(x - x_0)$, and **it is coprime to** $\frac{a}{d}$ (exercise 7), whence $\frac{b}{d} \mid (x - x_0)$. *(This step is the generalised Euclid lemma, and it follows from Bézout itself: from $u\frac{b}{d} + v\frac{a}{d} = 1$, multiplying by $x - x_0$, both summands turn out to be multiples of $\frac{b}{d}$.)* Hence $x - x_0 = k\frac{b}{d}$ for some k, and substituting, $y - y_0 = -k\frac{a}{d}$. ■

In Example 1.1.8 ($a = 252$, $b = 198$, $d = 18$): $\frac{b}{d} = 11$ and $\frac{a}{d} = 14$, so the family is $(4 + 11k,\ -5 - 14k)$. With $k = 1$: $15 \cdot 252 - 19 \cdot 198 = 3780 - 3762 = 18$ ✓.

> Non-uniqueness turns out to be harmless in every later use: in Session 3 **one** x with $ax \equiv 1 \pmod{n}$ is enough, and every member of the family gives **the same** inverse mod n — precisely because they differ by multiples of $\frac{b}{d}$.

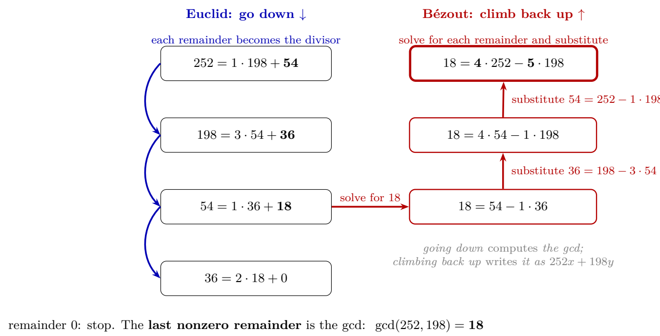

Figure 2: The mental picture of the extended algorithm, with Example 1.1.8: **Euclid goes down** (each remainder becomes the divisor, until the last nonzero remainder — the gcd) and **Bézout climbs back up** (solve for the gcd in the next-to-last division and substitute the remainders upwards, until it is written as a combination of the original inputs).

> **What Bézout is for (an announcement that will be honored):** in Session 2 it proves Euclid's lemma (the heart of unique factorization); in Session 3 it **decides and computes inverses** modulo n; in Session 4 it builds the solutions of the Chinese remainder theorem. It is, quite possibly, the most profitable result of the module.

Exercises

1. ★ Compute $\gcd(1001, 273)$ by the Euclidean algorithm, recording each division, and work back up to write it as $1001x + 273y$. Check numerically.
2. ★ Prove from the definition (choose and declare the **type of proof** — M0 connection): if $a \mid b$ and $b \mid a$, then $a = \pm b$.
3. ★ Find the quotient and remainder when dividing -100 by 7 (mind the convention $0 \le r < 7$!) and when dividing 100 by 7. What is the relation between the two remainders?
4. Justify from the definition: $\gcd(a, 0) = |a|$ (for $a \neq 0$) and $\gcd(a, 1) = 1$.
5. Prove that $\gcd(a, b) = \gcd(a, b - a)$ **without** using Euclidean division (only Prop. 1.1.2.2), and explain why this already "almost" justifies the Euclidean algorithm.
6. Give the uniqueness proof of Theorem 1.1.3 **by contradiction**, and compare it with the direct one in these notes (which do you find clearer? — a connection with M0).
7. Prove: if $\gcd(a, b) = d$, then $\gcd(a/d, b/d) = 1$. *(Hint: Bézout, divided by d.)*
8. The set $D = \{ax + by\}$ from the proof of Bézout: prove that D is exactly the set of **multiples** of d. *(The combinations of two numbers are the*

multiples of their gcd — the fine-grained version of Bézout.)

9. *(Characterization of the gcd by divisibility.)* Prove that $d = \gcd(a, b)$ satisfies: $d \mid a$, $d \mid b$, and **every** common divisor of a and b divides d. *(Hint: the last one follows from Bézout.)* Check also that those three conditions determine d **up to sign**, and that the convention $d \geq 0$ pins it down.

Going further and references

Going further (gcd of several numbers, and efficiency). $\gcd(a, b, c) = \gcd(\gcd(a, b), c)$, and the Euclidean algorithm is **genuinely fast**: the number of steps is proportional to the number of digits (the worst case is given by the Fibonacci numbers — Lamé's theorem). That efficiency, set against the cost of factoring, is the crack on which the cryptography of Session 4 is built. See Childs, *A Concrete Introduction to Higher Algebra*, ch. 3–4.

Going further (why the gcd is not computed by brute force). The direct route would be to list the divisors of each number, cross the two lists and keep the largest. With Example 1.1.8 it works: the divisors of 252 are $1, 2, 3, 4, 6, 7, 9, 12, 14, 18, 21, 28, 36, 42, 63, 84, 126, 252$; those of 198 are $1, 2, 3, 6, 9, 11, 18, 22, 33, 66, 99, 198$; the largest common one is 18.

But measure the work. To list the divisors of n by trial it is enough to test candidates up to $\sqrt{n}$ —divisors come in pairs, d with n/d, and one of the two is $\leq \sqrt{n}$—, so that is of the order of $\sqrt{n}$ trials per number. With three digits it is tedious; with twenty digits it is some 10^{10} trials per number, against the **four divisions** Euclid needed here and the few dozen it would need there.

And the contrast runs deeper: Euclid delivers the gcd **without factoring** either number. That matters because **no fast method for factoring** a large integer is known —an open problem, not a theorem— and RSA in Session 4 is built on that asymmetry: multiplying two large primes is immediate, recovering them from the product is not.

Going further (Euclidean domains). Any "world" with a reasonable division with remainder (a *Euclidean domain*) repeats this entire session word for word: gcd, the Euclidean algorithm, Bézout. The star example of the course will be $K[x]$ (Polynomials module); others, such as the Gaussian integers $\mathbb{Z}[i]$, in Aluffi, *Algebra: Chapter 0*, ch. V.

References. Childs, ch. 2–4; Dorronsoro–Hernández, *Números, grupos y anillos*, ch. 2.

Lecture Notes · Arithmetic · Session 2 — Primes and unique factorization

The session's complete proofs: the infinitude of the primes (with the preliminary lemma the classical proof usually takes for granted), Euclid's lemma via Bézout, and the fundamental theorem of arithmetic with existence **and uniqueness** in full detail. Exercises and a Going further section at the end.

2.1 Prime numbers

Definition 1.2.1. An integer $p > 1$ is **prime** if its only positive divisors are 1 and p. An integer > 1 that is not prime is **composite**. *(The number* 1 *is neither one thing nor the other — the reason for excluding it becomes visible in Theorem 1.2.6.)*

Lemma 1.2.2 (the least divisor is prime). Every integer $n > 1$ has some prime divisor; in fact, its **least** divisor greater than 1 is prime.

Proof. The set of divisors of n greater than 1 is nonempty (n is in it); let p be its minimum (well-ordering, Proposition 0.1.9). If p had a divisor d with $1 < d < p$, that d would divide n (transitivity, Prop. 1.1.2.1) and would be a divisor of n greater than 1 and smaller than p — against minimality. Hence p is prime. ∎

Theorem 1.2.3 (Euclid: infinitude of the primes). There are infinitely many prime numbers.

Proof (by contradiction — the most famous in mathematics). Suppose there were only finitely many: $p_1, p_2, \dots, p_k$. Consider

$$N = p_1 p_2 \cdots p_k + 1.$$

Since $N > 1$, it has a prime divisor p (Lemma 1.2.2) — which must be one from the list. But each p_i leaves **remainder** 1 when dividing N (it divides the product, not the 1; Prop. 1.1.2.2 with signs). Contradiction: p is a prime outside the list, which was supposed to be complete. ∎

> **M0 connection:** this is the canonical example of proof by contradiction (and a good place to recall, in passing, the remark on the excluded middle). Note also the role of Lemma 1.2.2: the "popular-science" version of this proof usually skips it.
>
> **The proof is constructive (no excluded middle?).** Although presented "by contradiction", at bottom it **exhibits** a new prime: given any finite list $p_1, \dots, p_k$, the number $N = p_1 \cdots p_k + 1$ has (Lemma 1.2.2) a prime divisor, and that prime is **not** in the list (each p_i leaves remainder 1). It is a procedure that, from known primes,

produces another — it does not require the excluded middle. The wording "suppose there are finitely many…" is a convenient wrapper, not a logical necessity. *(A nod to the constructivism of M0.)*

2.2 Euclid's lemma — where Bézout pays off

Lemma 1.2.4 (general version). If $\gcd(c, a) = 1$ and $c \mid ab$, then $c \mid b$.

Proof. Bézout (Theorem 1.1.9): $1 = cx + ay$. Multiplying by b:

$$b = cbx + aby.$$

c divides the first summand (obvious) and the second (because $c \mid ab$); hence it divides the sum (Prop. 1.1.2.2): $c \mid b$. ■

Corollary 1.2.5 (Euclid's lemma). If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof. If $p \nmid a$, then $\gcd(p, a) = 1$ (the positive divisors of p are 1 and p, and p does not divide a), and Lemma 1.2.4 gives $p \mid b$. ■

> **Structural counterfactual (the hypothesis is essential):** $6 \mid 4 \cdot 9 = 36$, but $6 \nmid 4$ and $6 \nmid 9$. Composites do not penetrate products: their ability to "split themselves" between the factors ($6 = 2 \cdot 3$, with the 2 going to the 4 and the 3 to the 9) is exactly what the lemma forbids the primes. Being prime is not just "not factoring": it is this property.

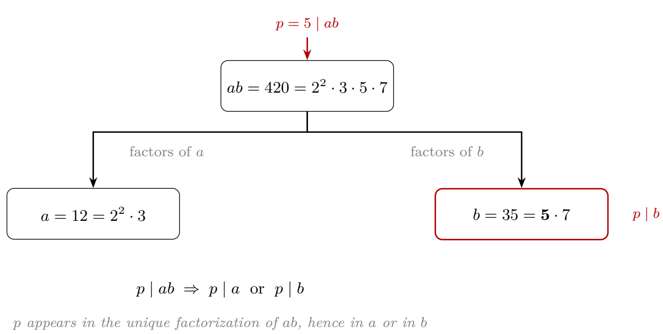

Figure 3: Via unique factorization, the primes of ab are those of a together with those of b; if $p \mid ab$, p is one of them, hence it is "in" a or "in" b. For a composite like 6, its two primes can split between the factors.

> **Heuristic, not proof.** It is a good mental image, but **circular as a proof** (the uniqueness of factorization is proved *with* this lemma): the rigorous argument is Bézout's (Lemma 1.2.4).

2.3 The fundamental theorem of arithmetic

Theorem 1.2.6 (fundamental theorem of arithmetic). Every integer $n > 1$ can be written as a product of primes, $n = p_1 p_2 \cdots p_r$, and the expression is **unique up to the order** of the factors.

Proof. **Existence (strong induction; see Preliminaries).** For $n = 2$: prime, a product of one factor. Let $n > 2$ and suppose the claim true for all integers between 2 and $n-1$. If n is prime, done. If not, n has a divisor a with $1 < a < n$ (Definition 1.2.1), and then $n = ab$ with $b = n/a$, which also satisfies $1 < b < n$: if $b = 1$ we would get $a = n$, and if $b = n$ we would get $a = 1$ — both cases excluded. By the induction hypothesis, a and b are products of primes, and concatenating, so is n. *(Here the **strong** version of induction is essential: a and b are smaller than n but not necessarily $n-1$, so ordinary induction — which only grants the case $n-1$ — would not suffice.)*

Uniqueness. Suppose $n = p_1 \cdots p_r = q_1 \cdots q_s$ with all the p_i, q_j prime; we argue by induction on r. Since $p_1 \mid q_1(q_2 \cdots q_s)$, Euclid's lemma (applied repeatedly) forces p_1 to divide **some** q_j; reordering, $p_1 \mid q_1$, and since both are prime, $p_1 = q_1$. Cancelling (n/p_1): $p_2 \cdots p_r = q_2 \cdots q_s$, a **smaller** number, to which the induction hypothesis applies: $r-1 = s-1$ and the factors coincide up to order. (Base case $r = 1$: $n = p_1$ prime admits no factorizations other than itself.) ■

Remark 1.2.7 (why 1 is not prime). If 1 counted as a prime, uniqueness would die on the spot: $6 = 2 \cdot 3 = 1 \cdot 2 \cdot 3 = 1 \cdot 1 \cdot 2 \cdot 3 = \cdots$ The exclusion is not a whim of nomenclature: it is the condition for Theorem 1.2.6 to be true as stated.

Remark 1.2.7b (negative integers and "units"). Theorem 1.2.6 is stated for $n > 1$ with positive primes, and then "up to order" is exact. If negative integers are allowed, uniqueness becomes "up to order **and a sign** ± 1": the ± 1 are the **units** of $\mathbb{Z}$ (its invertible elements), and the general statement of unique factorization — in any unique factorization domain — is "up to order and units". Note, in passing, that under Definition 1.2.1 **prime** and **composite** only classify the integers > 1: a negative number such as -4 is neither prime nor composite — its arithmetic is read through its positive part, $-4 = (-1) \cdot 2^2$, with the sign set aside as a unit — and 0 and ± 1 also fall outside the classification. Here the positive case suffices; the term "unit" is formalized in ring theory (references in the Going further section).

> *Structural lens:* $\mathbb{Z}$ has **atoms** (the primes) and unique composition. The complete skeleton — division with remainder → the Euclidean algorithm → Bézout → Euclid's lemma → unique factorization — will be rebuilt, piece by piece and with the same proofs, in $K[x]$ (Polynomials module). When it arrives, it will not be a new topic: it will be this one, in disguise.

2.4 Practical consequences

Proposition 1.2.8 (gcd and lcm via exponents). For nonzero a, b, writing $|a| = \prod p_i^{\alpha_i}$ and $|b| = \prod p_i^{\beta_i}$ (exponents ≥ 0, common primes), then

$$\gcd(a, b) = \prod p_i^{\min(\alpha_i, \beta_i)}, \qquad \operatorname{lcm}(a, b) = \prod p_i^{\max(\alpha_i, \beta_i)}, \qquad \gcd \cdot \operatorname{lcm} = |a\, b|.$$

(The sign of a, b is irrelevant for divisibility — hence the $|\cdot|$ and the $|ab|$, not ab, valid also with negatives. Proof: a divisor of a has, by the uniqueness in the fundamental theorem of arithmetic, exponents $\leq \alpha_i$; the rest is taking minima/maxima and the identity $\min + \max = \alpha + \beta$. *Details: exercise 5.)*

> **The convention is what makes the formula readable.** "The same primes p_i, with exponents ≥ 0" means: **both numbers are written over a single list of primes** — the union of the primes of a and those of b — putting **exponent** 0 wherever a prime does not occur. Without that convention the formula says nothing as soon as the two numbers share no factors: with $a = 4 = 2^2$ and $b = 9 = 3^2$, the common list is $\{2, 3\}$ and one writes $4 = 2^2 3^0$, $9 = 2^0 3^2$, whence $\gcd = 2^0 3^0 = 1$ and $\operatorname{lcm} = 2^2 3^2 = 36$.

> **Example 1.2.8b.** $a = 12$, $b = 90$. Common list of primes, $\{2, 3, 5\}$:
>
> $$12 = 2^2\, 3^1\, 5^0, \qquad 90 = 2^1\, 3^2\, 5^1.$$
>
> Minima of the exponents, $(1, 1, 0)$: $\gcd(12, 90) = 2 \cdot 3 = 6$. Maxima, $(2, 2, 1)$: $\operatorname{lcm}(12, 90) = 4 \cdot 9 \cdot 5 = 180$. *Check of the identity:* $6 \cdot 180 = 1080 = 12 \cdot 90$ ✓.

> **But beware the practical trap:** this formula requires **factoring**, and factoring is computationally expensive; the Euclidean algorithm computes the gcd **without factoring**, and blazingly fast. That asymmetry — multiplication/gcd cheap, factoring expensive — is not an anecdote: it is the basis of the cryptography of Session 4.

Proposition 1.2.9 (primality test up to $\sqrt{n}$). If $n > 1$ has no prime divisor $p \leq \sqrt{n}$, then n is prime.

Proof. If $n = ab$ with $1 < a \leq b < n$, then $a^2 \leq ab = n$, so $a \leq \sqrt{n}$, and the least prime divisor of a (Lemma 1.2.2) is a prime divisor of n less than or equal to $\sqrt{n}$. ■

Exercises

1. ★ Factor 360 and 756; compute the gcd and lcm via exponents, and **redo the gcd with the Euclidean algorithm**. Compare the work of the two methods and explain which would scale better with 100-digit numbers.
2. ★ **(Counterfactual.)** Find **your own** counterexample to Euclid's lemma with a **composite number** (different from $6 \mid 4 \cdot 9$) and point out how the composite "splits" between the factors.

3. ★ Is 221 prime? And 223? Decide with the test of Proposition 1.2.9, listing exactly which primes must be tried and why those suffice.
4. ★ Answer with the fundamental theorem of arithmetic: why is 1 not prime? (The correct answer is not "because the definition says so".)
5. Complete the proof of Proposition 1.2.8 (divisors via exponents; the $\min + \max$ identity).
6. Prove that $\sqrt{2}$ is irrational using the fundamental theorem of arithmetic: in $a^2 = 2b^2$, count the parity of the exponent of 2 on each side. *(Compare with the "classical" proof by contradiction from M0 — same conclusion, finer tool.)*
7. Prove that there are infinitely many primes of the form $4k+3$. *(Guide: adapt Euclid's proof with $N = 4p_1 \cdots p_k - 1$; you will need: a product of numbers of the form $4k+1$ is of the form $4k+1$.)*
8. If p is prime and $p \mid a^n$, prove that $p^n \mid a^n$. *(Iterated Euclid's lemma + fundamental theorem of arithmetic.)*

Going further and references

Going further (the distribution of the primes). Knowing there are infinitely many opens the question of *how many*: the **prime number theorem** says that up to x there are approximately $x/\ln x$. Between the arithmetic of this session and that result lie a century and a half of analysis. A readable overview: Childs, or ch. 1 of Hardy–Wright, *An Introduction to the Theory of Numbers.*

Going further (unique factorization that fails). There are "worlds" where the fundamental theorem of arithmetic is false: in $\mathbb{Z}[\sqrt{-5}]$, $6 = 2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$ are two genuinely different factorizations. Uniqueness is not free: it depends on Euclid's lemma, which depends on Bézout, which depends on division with remainder. Aluffi, ch. V.

References. Childs, ch. 5–6; Dorronsoro–Hernández, ch. 2–3.

Lecture Notes · Arithmetic · Session 3 — Congruences and $\mathbb{Z}/n\mathbb{Z}$

The heart of the module, with everything proved: the equivalence relation, the *well-definedness* of the quotient operations, the invertibility criterion (with the effective computation of inverses), the **zero divisor / non-invertible dichotomy** — which prepares the notion of a **field**, to be formalized later in the course —, $\mathbb{Z}/p\mathbb{Z}$ as a first field, and Fermat's little theorem with its rearrangement proof. Exercises and a Going further section at the end.

3.1 Congruences

Definition 1.3.1. Fix $n \geq 2$: $a \equiv b \pmod{n}$ if $n \mid (a - b)$.

Proposition 1.3.2. $a \equiv b \pmod{n} \iff a$ and b leave the **same remainder** when divided by n.

Proof. Writing $a = nq_1 + r_1$, $b = nq_2 + r_2$ with $0 \leq r_i < n$ (Theorem 1.1.3): $a - b = n(q_1 - q_2) + (r_1 - r_2)$. If $r_1 = r_2$, then $n \mid (a - b)$. Conversely, if $n \mid (a - b)$, then $n \mid (r_1 - r_2)$ (Prop. 1.1.2.2), and since $|r_1 - r_2| < n$, it must be that $r_1 = r_2$ (the bounding trick from Session 1). ■

Proposition 1.3.3. Congruence modulo n is an **equivalence relation** on $\mathbb{Z}$ (the concept from M0, at work!).

Proof. Reflexive: $n \mid 0$. *Symmetric:* if $n \mid (a - b)$ then $n \mid (b - a)$ (the negative). *Transitive:* if $n \mid (a - b)$ and $n \mid (b - c)$, then $n \mid ((a - b) + (b - c)) = (a - c)$ (Prop. 1.1.2.2). ■

Definition 1.3.4. The **class** of a is $\bar{a} = \{\, b : b \equiv a \pmod{n} \,\} = \{a, a \pm n, a \pm 2n, \ldots\}$. By Proposition 1.3.2 there are exactly n **classes**, one per remainder: $\bar{0}, \bar{1}, \ldots, \overline{n-1}$.

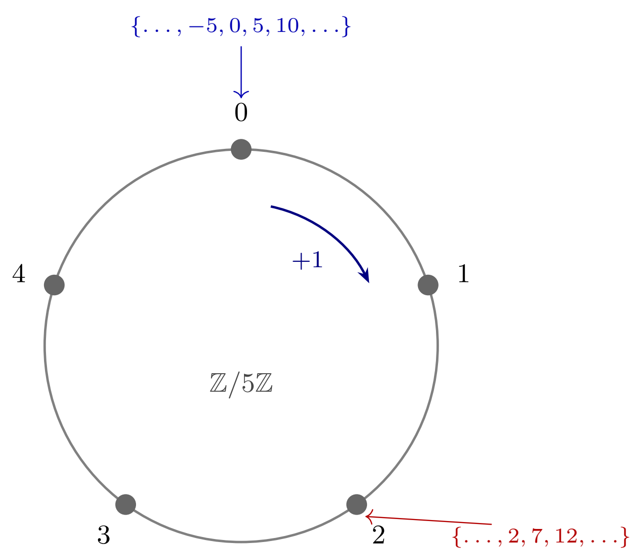

Figure 4: Congruence modulo n: wrap the number line around a circle with n marks; integers congruent modulo n land on the same mark (here $n = 5$).

3.2 $\mathbb{Z}/n\mathbb{Z}$: operating with classes (and the toll of the quotient)

Definition 1.3.5. $\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$, with

$$\bar{a} + \bar{b} := \overline{a+b}, \qquad \bar{a} \cdot \bar{b} := \overline{a \cdot b}.$$

Proposition 1.3.6 (the operations are well defined). If $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, then $a + b \equiv a' + b'$ and $ab \equiv a'b' \pmod{n}$.

Proof. Sum: $(a+b) - (a'+b') = (a-a') + (b-b')$, a sum of two multiples of n. Product: the trick of **adding and subtracting the cross term**:

$$ab - a'b' = ab - a'b + a'b - a'b' = (a-a')\,b + a'\,(b-b'),$$

a combination of multiples of n (Prop. 1.1.2.2). ■

> **Why this is THE fine point:** Definition 1.3.5 computes with **representatives** of the classes. If the result depended on the chosen representative, the definition would be nonsense. Proposition 1.3.6 is the **mandatory toll of every quotient construction** — a pattern worth internalizing here, in the most concrete example possible (M0: equivalence classes).

The properties of the operations (associative, commutative, distributive, identity elements $\bar{0}, \bar{1}$, negatives) are **inherited** from $\mathbb{Z}$ representative by representative. *Structural lens:* a set with these properties is called a **ring** — we drop the name in passing; its general theory is not part of this course.

3.3 Invertibles and zero divisors

Definition 1.3.7. $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ is **invertible** if there exists $\bar{x}$ with $\bar{a}\bar{x} = \bar{1}$. And $\bar{a} \neq \bar{0}$ is a **zero divisor** if there exists $\bar{b} \neq \bar{0}$ with $\bar{a}\bar{b} = \bar{0}$.

Theorem 1.3.8 (invertibility criterion). $\bar{a}$ is invertible in $\mathbb{Z}/n\mathbb{Z}$ $\iff$ $\gcd(a,n) = 1$ — that is, $\iff$ a and n are **coprime** (Definition 1.1.5: sharing no factor greater than 1). Moreover, in that case the inverse **is computed** with the extended Euclidean algorithm.

Proof. ($\Leftarrow$) Bézout: $1 = ax + ny$ for some x, y. Modulo n: $\overline{ax} = \bar{1}$, that is, $\bar{a}\bar{x} = \bar{1}$ — and the Bézout coefficients **are** the inverse. ($\Rightarrow$) If $\bar{a}\bar{x} = \bar{1}$, then $n \mid (ax-1)$, i.e. $ax - 1 = ny$: $1 = ax - ny$. Every common divisor of a and n divides that combination (Prop. 1.1.2.2), hence divides 1: the gcd is 1. ■

> **Example 1.3.9.** Inverse of $\bar{7}$ in $\mathbb{Z}/30\mathbb{Z}$: the Euclidean algorithm gives $30 = 4 \cdot 7 + 2$, $7 = 3 \cdot 2 + 1$; working back up, $1 = 7 - 3 \cdot 2 = 7 - 3(30 - 4 \cdot 7) = 13 \cdot 7 - 3 \cdot 30$. Hence $\bar{7}^{-1} = \overline{13}$. *Check:* $7 \cdot 13 = 91 = 3 \cdot 30 + 1$ ✓.

Proposition 1.3.9b (linear congruences, the general case). The congruence $ax \equiv b \pmod{n}$ has a solution **if and only if** $d = \gcd(a,n)$ divides b; and in that case it has **exactly** d **solutions** modulo n: if x_0 is the unique solution of $\frac{a}{d}\,x \equiv \frac{b}{d} \pmod{\frac{n}{d}}$ (unique because $\frac{a}{d}$ is invertible modulo $\frac{n}{d}$), the solutions are

$$x_0,\ x_0 + \tfrac{n}{d},\ x_0 + 2\tfrac{n}{d},\ \dots,\ x_0 + (d-1)\tfrac{n}{d} \pmod{n}.$$

Proof. Solving $ax \equiv b \pmod{n}$ is solving $ax + ny = b$ over the integers, which has a solution $\iff$ $d \mid b$ (it is Bézout's equation, scaled: Session 1). If $d \mid b$, dividing **everything** by d yields the equivalent congruence $\frac{a}{d}x \equiv \frac{b}{d} \pmod{\frac{n}{d}}$, where $\gcd(\frac{a}{d}, \frac{n}{d}) = 1$: by Theorem 1.3.8 there is a unique solution x_0 modulo $\frac{n}{d}$.

The count. Since the reduction is an **equivalence**, x is a solution $\iff$ $x \equiv x_0 \pmod{n/d}$ $\iff$ $x = x_0 + k\frac{n}{d}$ for some $k \in \mathbb{Z}$: that gives them all. It remains to see how many are **distinct modulo** n. Two of them coincide if and only if

$$n \,\big|\, (k-k')\tfrac{n}{d} \iff d \mid (k-k'),$$

(dividing by n/d), so k only counts **modulo** d: there are exactly d classes, those of $k = 0, 1, \dots, d-1$. Put the other way round, which is how one remembers it: on reaching $k = d$ one gets $x_0 + d\frac{n}{d} = x_0 + n \equiv x_0$ — the cycle closes and no new solution appears. ■

> **Example 1.3.9c.** $6x \equiv 9 \pmod{15}$: $d = \gcd(6, 15) = 3$ divides 9 ✓. Dividing by 3: $2x \equiv 3 \pmod{5}$, and since $\bar{2}^{-1} = \bar{3}$ in $\mathbb{Z}/5\mathbb{Z}$ $(2 \cdot 3 = 6 \equiv 1)$, $x \equiv 9 \equiv 4 \pmod{5}$. The **three** solutions modulo 15: $x \in \{4, 9, 14\}$. *Check:* $6 \cdot 9 = 54 = 3 \cdot 15 + 9$ ✓.

Theorem 1.3.10 (the dichotomy). Let $\bar{a} \neq \bar{0}$ in $\mathbb{Z}/n\mathbb{Z}$. Then:

$$\bar{a} \text{ is a zero divisor} \iff \bar{a} \text{ is not invertible} \iff \gcd(a, n) > 1.$$

Proof. The second equivalence is Theorem 1.3.8 negated. We prove the first in both directions.

(Zero divisor ⇒ not invertible.) Let $\bar{a}\bar{b} = \bar{0}$ with $\bar{b} \neq \bar{0}$. If $\bar{a}$ had an inverse, multiplying: $\bar{b} = (\bar{a}^{-1}\bar{a})\bar{b} = \bar{a}^{-1}(\bar{a}\bar{b}) = \bar{a}^{-1}\bar{0} = \bar{0}$ — contradiction.

(Not invertible ⇒ zero divisor.) Let $d = \gcd(a, n) > 1$ and take $\bar{b} = \overline{n/d}$. Then $\bar{b} \neq \bar{0}$ (because $0 < n/d < n$, since $d > 1$), and

$$a \cdot \frac{n}{d} = \frac{a}{d} \cdot n \equiv 0 \pmod{n}$$

(it is an integer multiple of n: $a/d \in \mathbb{Z}$ because $d \mid a$). Hence $\bar{a}\bar{b} = \bar{0}$: a zero divisor. ■

Remark 1.3.11 (the dichotomy is special to $\mathbb{Z}/n\mathbb{Z}$). In $\mathbb{Z}$, the number 2 is **not** invertible and is **not** a zero divisor either: outside the $\mathbb{Z}/n\mathbb{Z}$'s the two notions need not fill the whole space. *(This is exactly the caution that will reappear later, when the* ***fields*** *are presented in Linear Algebra: here it is already proved in the most concrete example.)*

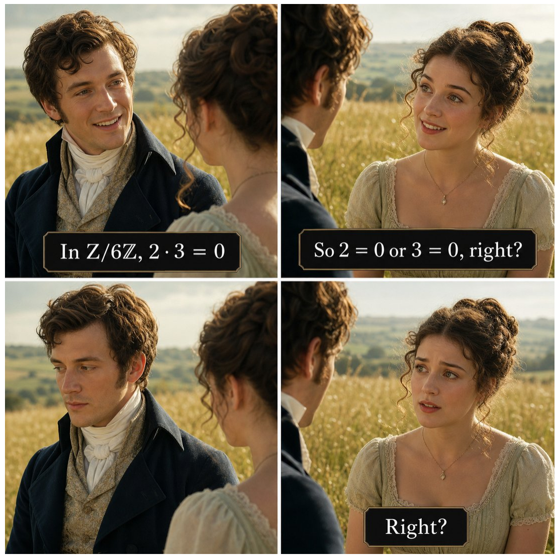

Figure 5: The reflex this section dismantles: in $\mathbb{Z}$ (or in $\mathbb{R}$!) a zero product betrays a zero factor, but in $\mathbb{Z}/6\mathbb{Z}$ we have $\bar{2} \cdot \bar{3} = \bar{0}$ with both factors nonzero — zero divisors, exactly what Theorem 1.3.10 detects via gcd > 1.

Corollary 1.3.12 (the thread of fields begins). 1. If p is **prime**, every nonzero element of $\mathbb{Z}/p\mathbb{Z}$ is invertible — a finite world where **one can divide**. It gets a *name*: $\mathbb{Z}/p\mathbb{Z}$ is a **field**. This is the start of the course's **thread of fields**: $\mathbb{Z}/p\mathbb{Z}$ here; second stop, $\mathbb{C}$, in the Complex Numbers module; and the formal definition and its theory, in the Linear Algebra module. 2. If n is **composite**, $\mathbb{Z}/n\mathbb{Z}$ has zero divisors and is **not** a field.

Proof. (1) $0 < a < p$ and p prime $\Rightarrow \gcd(a, p) = 1$ (the only divisors of p are 1 and p) $\Rightarrow$ invertible (1.3.8). (2) $n = ab$ with $1 < a, b < n$ gives $\bar{a}\bar{b} = \bar{0}$ with both nonzero. ∎

×	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

$\mathbb{Z}/4\mathbb{Z}$: there are zero divisors
$(2 \cdot 2 = 0,\ 2 \neq 0)$

×	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

$\mathbb{Z}/5\mathbb{Z}$: a field
(no product of nonzero elements is 0)

Figure 6: Multiplication tables of $\mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}/5\mathbb{Z}$: in $\mathbb{Z}/4\mathbb{Z}$ there are zero divisors $(2 \cdot 2 = 0)$; $\mathbb{Z}/5\mathbb{Z}$, since 5 is prime, is a field.

Software (RA7) — the tables, and hunting for zero divisors. Build the multiplication tables of $\mathbb{Z}/5\mathbb{Z}$ and $\mathbb{Z}/6\mathbb{Z}$ and locate in them what Theorem 1.3.10 says. The second part is the more instructive one: `pow(a, -1, n)` fails **exactly** when the inverse does not exist. Before running it, predict which pairs it will fail on.

```python
# Multiplication tables of Z/5Z and Z/6Z: hunt for the zero divisors
for n in (5, 6):
    print(f"Z/{n}Z:")
    for a in range(n):
        print(" ", [a*b % n for b in range(n)])
    print()

# inverses: pow(a, -1, n) - raises exactly when there is none (why?)
print("inverse of 3 mod 7:", pow(3, -1, 7))
try:
    pow(2, -1, 6)
except ValueError as e:
    print("inverse of 2 mod 6:", e)
```

3.4 Fermat's little theorem

Theorem 1.3.13 (Fermat). If p is prime and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof (the rearrangement). Consider the nonzero classes $\bar{1}, \bar{2}, \dots, \overline{p-1}$ and multiply them all by $\bar{a}$:

$$\bar{a} \cdot \bar{1}, \ \ \bar{a} \cdot \bar{2}, \ \ \dots, \ \ \bar{a} \cdot \overline{p-1}.$$

Claim: this list is the same as before, rearranged. Indeed, the map $\bar{x} \mapsto \bar{a}\bar{x}$ is **injective** ($\bar{a}\bar{x} = \bar{a}\bar{y} \Rightarrow \bar{x} = \bar{y}$, multiplying by $\bar{a}^{-1}$, which exists by Corollary 1.3.12.1) and never produces zero ($\bar{a}\bar{x} = \bar{0}$ with $\bar{a}$ invertible would force $\bar{x} = \bar{0}$); an injective map from a **finite** set to itself is bijective (M0). Equating the products of the two lists:

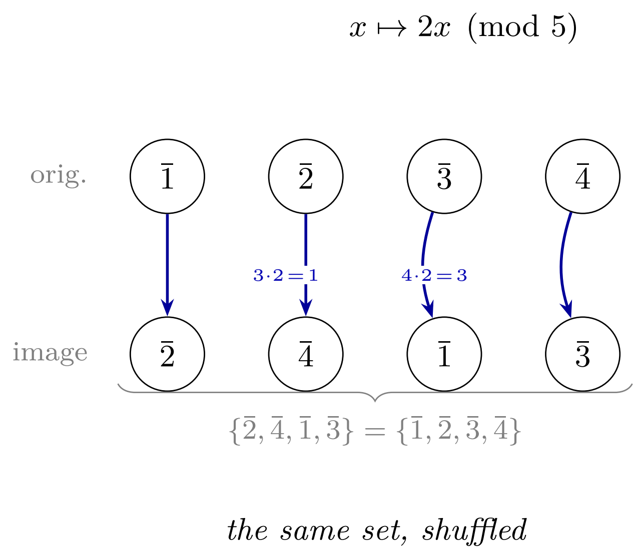

Figure 7: For $p = 5$, $a = 2$: the row $(\bar{1}, \bar{2}, \bar{3}, \bar{4})$ is transformed into $(\bar{2}, \bar{4}, \bar{1}, \bar{3})$ by $x \mapsto 2x$ ($1 \to 2$, $2 \to 4$, $3 \to 1$, $4 \to 3$) — the same set, shuffled.

That is why the product of the whole row does not change, and that is what gets cancelled.

$$\overline{(p-1)!} = \overline{a^{p-1}} \cdot \overline{(p-1)!}.$$

And $\overline{(p-1)!}$ is invertible (a product of invertibles); cancelling it: $\overline{a^{p-1}} = \bar{1}$. ■

Remark 1.3.13b (the temptation of the cycles, and why it fails). Looking at the rearrangement it is natural to ask how many times one must multiply by $\bar{a}$ to return to the starting point, and to expect the answer $p - 1$ — which would give $\bar{a}^{\,p-1} = \bar{1}$ for free. **It does not work that way.** That number is the **order** of $\bar{a}$, and although it always *divides* $p - 1$, it seldom equals it: for $p = 7$, $a = 2$ the cycle is $1 \to 2 \to 4 \to 1$, of length 3 and not 6; for $p = 11$, $a = 3$ it has length 5 and not 10. It coincides exactly when $\bar{a}$ is a **primitive root** modulo p (see Going further). The proof above never uses the cycles: it only uses that the arrival list has **the same elements** as the starting one. *(And the fact that the order divides $p - 1$ follows* ***from*** *Fermat, not the other way round: using it here would be circular.)*

> Note what the proof uses: that $\mathbb{Z}/p\mathbb{Z}$ is a **field** (all nonzero elements invertible, twice) and a **counting** argument (injective on a finite set = bijective, M0). **What it is for:** Fermat powers the modular exponentiation that sustains public-key cryptography — Session 4 collects the debt.

Software (RA7) — Fermat checked, and the nines trick. The first part verifies Theorem 1.3.13 with a large prime. The second is the same theorem in disguise: that $10 \equiv 1 \pmod 9$ is what makes the casting-out-nines check work.

```python
# Fermat's little theorem, checked; and the nines trick
p = 1009
print("2^(p-1) mod p =", pow(2, p-1, p), "(p prime)")

# The nines trick: n - digit_sum(n) is always a multiple of 9
n = 7345
rest = n - sum(int(c) for c in str(n))
print(n, "->", rest, "| multiple of 9?", rest % 9 == 0)
```

Exercises

1. ★ In $\mathbb{Z}/30\mathbb{Z}$, compute $\overline{17}^{-1}$ with the extended algorithm (not by trial and error) and check.
2. ★ **(Linear congruences, Proposition 1.3.9b.)** Solve $25x \equiv 15 \pmod{40}$: check the compatibility criterion first, say **how many** solutions there will be before computing any, and give them all modulo 40. Then explain why $25x \equiv 3 \pmod{40}$ has none. *(Careful with the division step: the modulus is divided too.)*
3. ★ Classify **all** the nonzero elements of $\mathbb{Z}/12\mathbb{Z}$ into invertibles and zero divisors; verify the dichotomy of Theorem 1.3.10 and exhibit, for each zero divisor, the partner $\bar{b}$ that betrays it.
4. ★ Compute the remainder of 3^{100} when divided by 7 (Fermat: $3^6 \equiv 1$) and that of 2^{1000} when divided by 11.
5. ★ **(The Linear Algebra counterfactual, now with theory.)** In $\mathbb{Z}/6\mathbb{Z}$: $\bar{2} \cdot \bar{3} = \bar{0}$. Locate the gcd that explains it via Theorem 1.3.10, and construct the "canonical" zero divisor $\overline{n/d}$ for $\bar{a} = \bar{2}$.
6. Prove that $\mathbb{Z}$ has no zero divisors, and conclude that the dichotomy 3.10 **fails** in $\mathbb{Z}$ (with 2 as the witness).
7. Prove the **divisibility rule for 9**: a number is divisible by 9 if and only if the sum of its digits is. *(Hint: $10 \equiv 1 \pmod 9$, so $10^k \equiv 1$.)*
8. Is it true that $a \equiv b \pmod n$ implies $a^2 \equiv b^2 \pmod n$? And the converse? Prove or refute (minimal counterexample).
9. **(Python, asynchronous activity.)** Generate the multiplication tables of $\mathbb{Z}/5\mathbb{Z}$ and $\mathbb{Z}/6\mathbb{Z}$ (`[[a*b % n for b in range(n)] for a in range(n)]`), spot the zero divisors visually, compute inverses with `pow(a, -1, n)` (built in; it raises an error exactly when no inverse exists — why?), and check Fermat with `pow(a, p-1, p)` for several primes.

Going further and references

Optional going further (rings). A set with two operations $+, \cdot$ satisfying the properties inherited here (both associative, addition commutative, distributive, with identity elements $\bar{0}, \bar{1}$ and negatives) is a **commutative ring with identity**; $\mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$ and (Polynomials module) $K[x]$ are examples. The **zero divisors** and the **units** (invertibles) of this session are general notions of ring theory; a commutative ring without zero divisors is an **integral domain**, and one where every nonzero element is invertible is a **field** (M4). To expand, optional: Dorronsoro–Hernández, *Números, grupos y anillos*, ch. 3; Childs, ch. 7–9; with a categorical flavor, Aluffi, *Algebra: Chapter 0*, ch. III. Not assessed.

Going further (Euler's φ function). How many invertibles does $\mathbb{Z}/n\mathbb{Z}$ have? Exactly $\varphi(n) = \#\{1 \le a \le n : \gcd(a, n) = 1\}$. **Euler's** theorem generalizes Fermat's: $a^{\varphi(n)} \equiv 1 \pmod n$ if $\gcd(a, n) = 1$ — and the same rearrangement proof works, restricted to the invertibles. For $n = pq$ (the RSA case): $\varphi(pq) = (p-1)(q-1)$. Childs, ch. 9; Dorronsoro–Hernández.

References. Childs, ch. 7–9 (congruences, $\mathbb{Z}/n\mathbb{Z}$, Fermat); Dorronsoro–Hernández, ch. 3. The dichotomy and fields: Linear Algebra notes, Session 1 (where what is proved here gets used).

Lecture Notes · Arithmetic · Session 4 — The Chinese remainder theorem and cryptography (RSA)

The module's closing session with everything proved: the constructive CRT (with the uniqueness lemma), fast modular exponentiation with the why behind it, and the theorem that **RSA decrypts** — Fermat and the CRT glued together, with the toy example verified by hand and the code of the demo. Exercises and a Going further section at the end.

4.1 The Chinese remainder theorem

Theorem 1.4.1 (CRT). If $\gcd(m, n) = 1$, the system of congruences

$$x \equiv a \pmod m, \qquad x \equiv b \pmod n$$

has a solution, and it is **unique modulo** mn (any two solutions are congruent modulo mn).

Proof. **Existence (constructive: the switches).** Bézout: $1 = mx_0 + ny_0$. Define

$$e_1 = ny_0, \qquad e_2 = mx_0.$$

Then $e_1 + e_2 = 1$, and: $e_1 \equiv 1 \pmod{m}$ (since $e_1 = 1 - mx_0$) and $e_1 \equiv 0 \pmod{n}$; symmetrically $e_2 \equiv 0 \pmod{m}$, $\equiv 1 \pmod{n}$. They are **switches**: each is worth 1 in one modulus and 0 in the other. The combination

$$x = a\,e_1 + b\,e_2$$

satisfies $x \equiv a \cdot 1 + b \cdot 0 = a \pmod{m}$ and $x \equiv b \pmod{n}$. ∎ *(existence)*

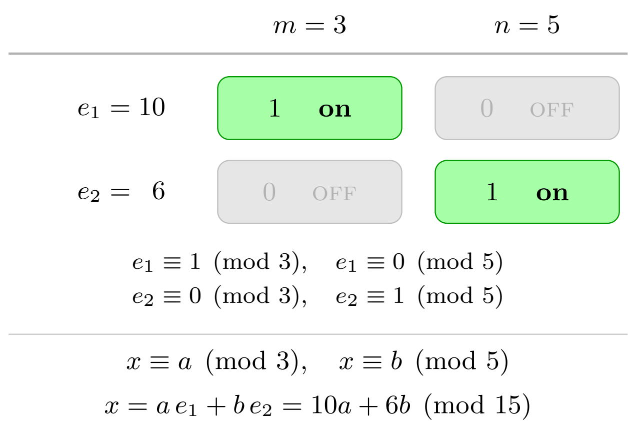

Figure 8: The CRT "switches": e_1 is worth 1 modulo m and 0 modulo n, and e_2 the other way around; combining them, $x = a\,e_1 + b\,e_2$ solves the system.

Lemma 1.4.2 (for the uniqueness). If $m \mid x$, $n \mid x$ and $\gcd(m, n) = 1$, then $mn \mid x$.

Proof. $x = mt$; since $n \mid mt$ and $\gcd(n, m) = 1$, Lemma 1.2.4 gives $n \mid t$, i.e. $t = ns$: $x = mns$. ∎

Uniqueness of the CRT. If x and x' solve the system, m and n divide $x - x'$, and by Lemma 1.4.2, $mn \mid (x - x')$. ∎

Corollary 1.4.2b (CRT for k moduli). If $m_1, \ldots, m_k$ are **pairwise coprime**, the system $x \equiv a_i \pmod{m_i}$ $(i = 1, \ldots, k)$ has a solution, unique modulo $m_1 m_2 \cdots m_k$.

Proof. Induction on k. The case $k = 2$ is Theorem 1.4.1. For the step, $M = m_1 \cdots m_{k-1}$ is coprime with m_k (a prime of m_k divides no m_i, $i < k$, by pairwise coprimality; fundamental theorem of arithmetic): solve the first $k - 1$ congruences by hypothesis (one congruence modulo M) and combine it with the

k-th one via the two-moduli case. *(Switch version:* $e_i \equiv 1 \pmod{m_i}$ *and* $\equiv 0$ *in the others, built with* $\widehat{m}_i = \prod_{j \neq i} m_j$*, which is invertible modulo* m_i*;* $x = \sum_i a_i e_i$*. Exercise 5.)* ■

> **Example 1.4.3 (the riddle).** $x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$. Bézout for $(3, 5)$: $1 = 2 \cdot 3 - 1 \cdot 5$, so $e_1 = -5$ ($\equiv 1 \bmod 3$, $\equiv 0 \bmod 5$) and $e_2 = 6$. Solution: $x = 2(-5) + 3(6) = 8$, unique modulo 15: $x \in \{8, 23, 38, \dots\}$. *Check:* $8 = 2 + 2 \cdot 3$ ✓, $8 = 3 + 5$ ✓.

Proposition 1.4.2c (arbitrary moduli: when there is a solution, and modulo what it is unique). Let $m, n \geq 1$ and $d = \gcd(m, n)$. The system $x \equiv a \pmod m$, $x \equiv b \pmod n$ has a solution **if and only if** $d \mid (b - a)$; and in that case the solution is **unique modulo** $\operatorname{lcm}(m, n)$.

Proof. **Existence.** The first congruence says exactly $x = a + ms$ with $s \in \mathbb{Z}$. Substituting into the second, $a + ms \equiv b \pmod n$, that is

$$m\, s \equiv b - a \pmod n.$$

This is a linear congruence in s, and **Proposition 1.3.9b** settles it: it has a solution $\iff$ $\gcd(m, n) = d$ divides $b - a$. Every such s yields an $x = a + ms$ solving the system, and conversely.

Uniqueness. If x and x' both solve it, then $m \mid (x - x')$ and $n \mid (x - x')$: the difference is a **common multiple** of m and n. And every common multiple is a multiple of the lcm — for $x = x'$ this is trivial, and if $D = x - x' \neq 0$ it is read off the exponents of **Proposition 1.2.8**: $m \mid D$ and $n \mid D$ force the exponent of each prime in D to be $\geq \alpha_i$ and $\geq \beta_i$, hence $\geq \max(\alpha_i, \beta_i)$. Conversely, adding $\operatorname{lcm}(m, n)$ to a solution yields another. The solutions are therefore exactly one class modulo $\operatorname{lcm}(m, n)$. ■

Theorem 1.4.1 is the case $d = 1$ of this. When m and n are coprime, $d = 1$ divides any $b - a$ — hence a solution *always* exists — and $\operatorname{lcm}(m, n) = mn$ by Proposition 1.2.8. Both claims of the CRT follow as a special case; what the CRT adds is the **construction** with switches, which is absent here.

The coprimality hypothesis is not decorative.

- *Unsolvable:* $x \equiv 0 \pmod 2$, $x \equiv 1 \pmod 4$: $\gcd(2, 4) = 2 \nmid (1 - 0) = 1$ — no solution (x even and at the same time $\equiv 1 \bmod 4$, impossible).
- *Solvable despite non-coprimality:* $x \equiv 2 \pmod 4$, $x \equiv 4 \pmod 6$: $\gcd(4, 6) = 2 \mid (4 - 2) = 2$ — there is a solution, unique modulo $\operatorname{lcm}(4, 6) = 12$. Trying: $x = 10$ satisfies $10 \equiv 2 \pmod 4$ and $10 \equiv 4 \pmod 6$; the solutions are $x \equiv 10 \pmod{12}$.

4.2 Fast modular exponentiation

These are **two distinct reductions**, and they should not be conflated: the first makes the computation *possible*, the second makes it *cheap*.

(1) Reduce at every multiplication. Instead of computing a^k and reducing at the end, reduce modulo n after each product. The factors are then always $< n$, so no intermediate result exceeds n^2: one never writes down a^k, which would have an astronomical number of digits. This already makes the computation fit in memory — but it still costs $k-1$ multiplications, hopeless if k has hundreds of digits.

(2) Square repeatedly. Each squaring **doubles the exponent**:

$$a^1 \;\to\; a^2 \;\to\; a^4 \;\to\; a^8 \;\to\; a^{16} \;\to\; \cdots$$

so that t multiplications reach exponent 2^t, not exponent t. To exploit this one must write the target exponent as a **sum of those powers of two**, and that is always possible, by a greedy rule: *take the largest power of* 2 *that fits and subtract, until nothing is left.* For $k = 13$: 8 fits, 5 left; 4 fits, 1 left; 1 fits, 0 left. Hence

$$13 = 8 + 4 + 1 \quad\Longrightarrow\quad a^{13} = a^8 \cdot a^4 \cdot a^1.$$

> **That such a decomposition always exists** is an exercise in strong induction (exercise 7), and the list of powers used and unused is precisely the **binary expansion** of k — which is why the cost of this algorithm is measured in *bits* of the exponent. But the procedure needs no knowledge of binary: the greedy rule above is all of it.

> **Example 1.4.4.** $3^{13} \bmod 7$: $3^2 = 9 \equiv 2$; $3^4 \equiv 2^2 = 4$; $3^8 \equiv 4^2 = 16 \equiv 2$. Then $3^{13} = 3^8 \cdot 3^4 \cdot 3 \equiv 2 \cdot 4 \cdot 3 = 24 \equiv 3 \pmod 7$.

Cost: on the order of the **number of bits of** k — about $2\log_2 k$ multiplications, and $\log_2 k$ is precisely that number of bits —: for a thousand-bit k, **a few thousand** operations, not 2^{1000}. Without this algorithm, RSA would be theory without practice.

4.3 RSA

What it solves: public-key cryptography

Before the scheme, the problem. To **encrypt** a message is to transform it with a key so that only whoever holds the right key can recover it. *Classical* (symmetric) cryptography uses the **same** secret key to encrypt and to decrypt: sender and receiver must **share it beforehand** over a secure channel — but if they already had a secure channel, a good part of the original problem would already be solved. That vicious circle — the **key-exchange problem** — limited cryptography for centuries.

The idea of **public-key** cryptography breaks it. Each user has a mathematically linked **pair** of keys: a **public** one, published openly, which anyone uses to encrypt messages for them, and a **private** one, kept secret, which is the only one capable of decrypting. No secret ever needs to be shared in advance. **RSA** (Rivest–Shamir–Adleman, 1978) was the first practical system to realize this idea, and it still protects — among many other things — the web connections with a "padlock". What keeps the public key from betraying the private one is a **computational asymmetry** (multiplying is easy, factoring is not), which we make precise in *"Why it is secure"*.

The scheme

1. **Keys.** Choose primes $p \neq q$ (large) — the distinctness is a **hypothesis of Theorem 1.4.5**, not a precaution: with $p = q$ the number $(p-1)(q-1)$ is no longer the one that makes decryption work (for $n = p^2$ the correct exponent is $p(p-1)$), and besides, the proof uses $\gcd(p, q) = 1$; $n = pq$; $\varphi = (p-1)(q-1)$; choose e with $\gcd(e, \varphi) = 1$, and compute $d = e^{-1} \bmod \varphi$ — **the extended algorithm from S1/S3, once again**. **Public** key: (n, e). **Private** key: d.
2. **Encrypt** (anyone): $c = m^e \bmod n$. **Decrypt** (only whoever holds d): $m = c^d \bmod n$. (Messages $0 \leq m < n$.)

Theorem 1.4.5 (RSA decrypts). With the keys built this way, for every m: $(m^e)^d \equiv m \pmod{n}$.

Proof. Since $ed \equiv 1 \pmod{\varphi}$, we write $ed = 1 + k(p-1)(q-1)$. **We work modulo p and modulo q separately, and glue with the CRT.**

Modulo p: if $p \mid m$, both sides are $\equiv 0$. If $p \nmid m$, Fermat (Theorem 1.3.13) gives $m^{p-1} \equiv 1$, so

$$m^{ed} = m \cdot \left(m^{p-1}\right)^{k(q-1)} \equiv m \cdot 1 = m \pmod{p}.$$

Modulo q: identical. Thus p and q both divide $m^{ed} - m$; since $\gcd(p, q) = 1$, **Lemma 1.4.2** gives $pq \mid (m^{ed} - m)$, that is, $m^{ed} \equiv m \pmod{n}$. ■

> Inventory of the proof: **Bézout** (built d), **Fermat** (the engine), **CRT/Lemma 1.4.2** (the glue). The entire module, in a ten-line proof.

Why it is secure (honesty)

To decrypt one needs d; to compute d one needs $\varphi = (p-1)(q-1)$; and knowing φ together with n **is equivalent to factoring** n (exercise 10: from n and φ one recovers p and q by solving a quadratic). Security rests, then, on the **computational asymmetry** of Session 2: multiplying is instantaneous; factoring a 2048-bit n, infeasible with known technology.

Multiplying versus factoring (the asymmetry, with intuition). It is worth seeing why one direction is easy and the other is not. **Multiplying** two large numbers is fast: the work grows *smoothly* with the number of **digits** — the school algorithm uses on the order of (number of digits)^2 operations; one says it is **polynomial** in the size of the input (the “size” is the number of digits, not the value of the number). For the reverse path — given n, **finding its divisors** — no fast method is known: in essence one must **search** among candidates, and that search space grows **explosively** (exponentially) with the number of digits. The gap becomes abysmal as soon as the numbers are large: with two primes of about 300 digits each (≈1024 bits), multiplying them is instantaneous, while factoring the ~600-digit product would exceed the age of the universe using all of today’s computers at once. *That* distance — trivial in one direction, intractable in the other — is the security of RSA. And it is robust against hardware progress: if the machines improve, **enlarging the keys** makes encryption slightly more expensive and factoring vastly more so.

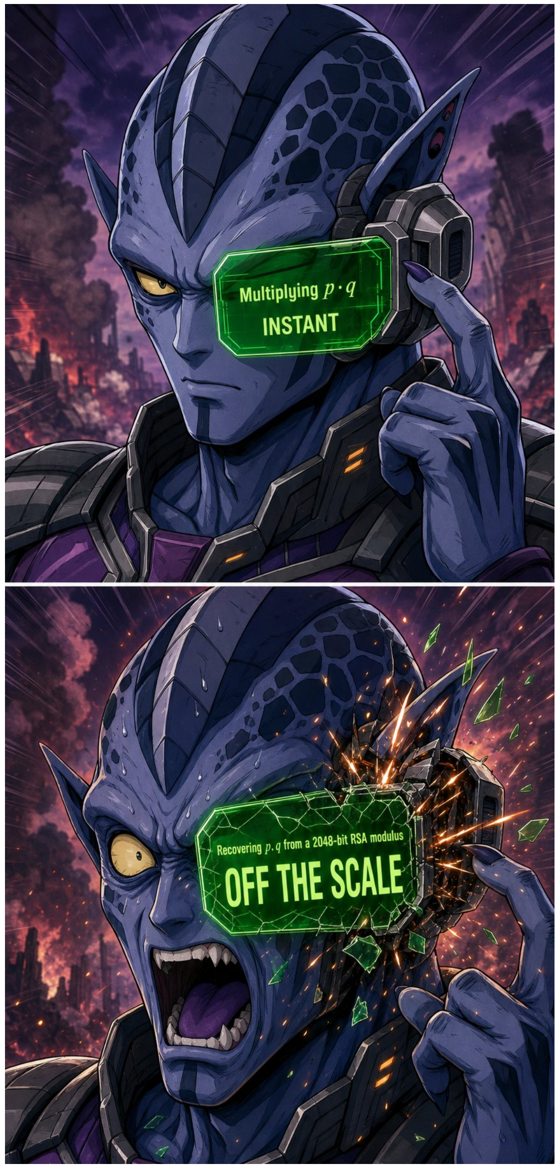

Figure 9: The asymmetry, in one image: the forward task barely registers on the meter; the inverse one blows it up.

Note (quantum computing). "Infeasible" refers to **classical** computers. In 1994 Peter Shor discovered an algorithm that factors **efficiently**… on a **quantum** computer. That RSA still stands today only means that quantum machines large enough to factor real keys do not yet exist. That is why **post-quantum cryptography** — based on problems conjectured to be hard even for quantum computers — is already being standardized. (Going further; cf. Childs.)

(Honest fine print: "equivalent de facto" — real-world RSA also carries random padding and implementation precautions without which there are attacks that do not require factoring. Ours is the mathematical core.)

Sufficient, not provably necessary. Read the chain in the direction in which it is proved: **factoring** n **suffices** to recover φ, hence d, and with d to read any message. The converse — that breaking RSA **requires** factoring — is an **open problem**: no algorithm is known that breaks RSA without factoring, but neither is there a proof that none exists. Saying "to break it one needs to factor" therefore claims more than is known; the correct statement is: *all the known security rests on the hardness of factoring.*

Example 1.4.6 (a toy one, verified by hand)

$p = 3$, $q = 11$: $n = 33$, $\varphi = 20$. We take $e = 3$ ($\gcd(3, 20) = 1$); the extended algorithm gives $d = 7$ ($3 \cdot 7 = 21 \equiv 1 \bmod 20$). **Encrypt** $m = 5$: $c = 5^3 = 125 \equiv 26 \pmod{33}$. **Decrypt:** $26^7 \bmod 33$, by squaring ($26 \equiv -7$): $(-7)^2 = 49 \equiv 16$; $(-7)^4 \equiv 16^2 = 256 \equiv 25$; $26^7 \equiv 25 \cdot 16 \cdot (-7) \equiv 4 \cdot (-7) = -28 \equiv \mathbf{5} \pmod{33}$ ✓ — the message comes back.

The Python demo (skeleton)

Everything is standard Python except `nextprime`, which comes from `sympy` (free: `pip install sympy`); modular exponentiation and the modular inverse are **built in** (three-argument `pow` and `pow(e, -1, phi)`).

```
from sympy import nextprime
from math import gcd

p = nextprime(10**6); q = nextprime(2 * 10**6)
n = p*q; phi = (p-1)*(q-1)
e = 65537; assert gcd(e, phi) == 1
d = pow(e, -1, phi)       # modular inverse (Bézout under the hood)
m = 271828                # the "message"
c = pow(m, e, n)          # encrypt (public key)
assert pow(c, d, n) == m  # decrypt (private key) - m comes back
```

*(This block is the starting point of the **SE3 mini-project**: you generate your keys, publish the public one on the course forum, and exchange signed messages with the rest of the class. The **full statement**, in SE3 · RSA mini-project. The **submission** goes through the virtual classroom.)*

Exercises

1. ★ Solve with the switches: $x \equiv 3 \pmod 7$, $x \equiv 5 \pmod{11}$. Give the general solution and the least positive one; check.
2. ★ RSA by hand with $p = 5$, $q = 11$, $e = 3$: build n, φ, d; encrypt $m = 4$ and decrypt the result, all by fast exponentiation.
3. ★ Compute $7^{45} \bmod 11$ combining **Fermat** (reduce the exponent) and successive squaring.
4. ★ **(Python — SE3 warm-up.)** Set up the full scheme with primes of 6–8 digits; encrypt a numeric message and verify the decryption. This is the starting point of the mini-project: get there with this already done. *(Breaking someone else's RSA by factoring is phase 5 of the project, with its ladder of targets.)*
5. Complete the **switch construction** of Corollary 1.4.2b: for pairwise coprime $m_1, \dots, m_k$, define $\widehat{m}_i = \prod_{j \neq i} m_j$, find its inverse modulo m_i, form the e_i and verify that $x = \sum_i a_i e_i$ solves the system. Apply it to $x \equiv 2\,(3),\ x \equiv 3\,(5),\ x \equiv 2\,(7)$.
6. **(Counterfactual completed.)** Redo Proposition 1.4.2c by hand, with **non**-coprime moduli. (a) $x \equiv 3 \pmod{12}$, $x \equiv 9 \pmod{18}$: check the criterion, find the solution by reducing to a linear congruence in s, and verify that the solutions form **a single class modulo** $\operatorname{lcm}(12, 18) = 36$ — not modulo $12 \cdot 18 = 216$. (b) $x \equiv 3 \pmod{12}$, $x \equiv 7 \pmod{18}$: what does the criterion say, and why was it foreseeable without computing anything? (c) **Prove** that, in general, if the system has a solution then the solution set is exactly one class modulo $\operatorname{lcm}(m, n)$ — that is, reproduce the uniqueness part of 1.4.2c without looking at it.
7. ★ **(The decomposition behind fast exponentiation.)** Prove by **strong induction** that every integer $k \geq 1$ is a sum of **distinct** powers of 2, and that the greedy rule of §4.2 — take the largest that fits and subtract — produces it. *(Hint: if $2^t \leq k < 2^{t+1}$, apply the hypothesis to $k - 2^t$ and check that this remainder is $< 2^t$, so no power is repeated.)* Deduce how many multiplications $a^k \bmod n$ costs by that method, in terms of the number of binary digits of k.
8. Why does RSA require $\gcd(e, \varphi) = 1$? Point out the step of the construction that would break without it.
9. **(Where does 65537 come from?)** The code in §4.3 fixes $e = 65537$ without justifying it. (a) Prove that e must be **odd**. (b) **(Python.)** Write the loop running through $e = 3, 5, 7, \dots$ until $\gcd(e, \varphi) = 1$, and count how many candidates it tries before succeeding, with 6-digit primes; repeat with several pairs p, q. Is "brute force" a problem here? (c) The other way round: if you **fix** $e = 65537$ *before* generating p and q, and use that 65537 is prime, what single condition must p and q satisfy? How expensive is checking it, compared with the loop in (b)? — And why *that* prime: write $65537 = 2^{16} + 1$ in binary and apply the count of **exercise 7**. Compare it with an arbitrary 17-bit e. *($e = 3$ would cost only 2 multiplications: why do you think it is not used? Cf. the fine print in §4.3.)*

10. Prove that knowing $n = pq$ and $\varphi = (p-1)(q-1)$ allows one to recover p and q. *(Hint: $p+q = n-\varphi+1$; then p, q are the roots of $t^2 - (p+q)t + n$.)*

Going further and references

Going further (from Fermat to Euler, and general RSA). Our Theorem 1.4.5 avoids the general φ function by using Fermat + CRT. The alternative route: **Euler's** theorem ($a^{\varphi(n)} \equiv 1$ if $\gcd(a, n) = 1$; Going further of Session 3) gives the case $\gcd(m, n) = 1$ in one line — but it does not cover the messages not coprime with n, which our proof does. Childs, ch. 9–10.

Going further (digital signatures). Swapping the roles of the keys (d to "encrypt", e to verify), RSA produces **signatures**: only the owner of d could have generated the signature, and anyone can verify it with the public key. The same mathematics, read backwards. Childs, ch. 10.

References. CRT: Childs, ch. 8; Dorronsoro–Hernández. RSA: Childs, ch. 10 (the reference closest to the course); the original Rivest–Shamir–Adleman paper (1978) is surprisingly readable.